



$$\lim_{x \rightarrow 1} \frac{(x^2 - 1)(\sqrt{x} + 1)}{(x - 1)} = \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x+1)(\sqrt{x}+1)}{\cancel{(x-1)}}$$

$$\lim_{x \rightarrow 1} (x+1)(\sqrt{x}+1) = 4$$

$$\lim_{x \rightarrow 4} \frac{(2 - \sqrt{x})(3 + \sqrt{x+5})(2 + \sqrt{x})}{(4 - x)(2 + \sqrt{x})} =$$

$$\lim_{x \rightarrow 4} \frac{\cancel{(4-x)}(3 + \sqrt{x+5})}{\cancel{(4-x)}(2 + \sqrt{x})} =$$

$$= \lim_{x \rightarrow 4} \frac{3 + \sqrt{x+5}}{2 + \sqrt{x}} = \lim_{x \rightarrow 4} \frac{3 + \sqrt{9}}{2 + \sqrt{4}} =$$

$$\frac{6}{4} = \frac{3}{2} = 1,5$$

$$\lim_{x \rightarrow 8} \frac{x-8}{\sqrt[3]{x}-2} = \frac{x^{1/3} - 8^{1/3}}{(x^{2/3} + x^{1/3} \cdot 8^{1/3} + 8^{2/3})}$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$\begin{array}{r} (x-8) \\ x^{\frac{2}{3}} + \cancel{x^{\frac{1}{3}} \cdot 8^{\frac{1}{3}}} + \cancel{x^{\frac{1}{3}} \cdot 8^{\frac{2}{3}}} - \cancel{8^{\frac{1}{3}} \cdot x^{\frac{2}{3}}} \\ \hline - \cancel{x^{\frac{1}{3}} \cdot 8^{\frac{1}{3}}} - 8^{\frac{1}{3}} \cdot 8^{\frac{2}{3}} = \end{array}$$

$$x^{\frac{2}{3}} - 8^{\frac{2}{3}} = \boxed{x - 8}$$

$$\lim_{x \rightarrow 8} \frac{\cancel{(x-8)}}{\cancel{(x-8)}} \cdot \frac{(x^{\frac{2}{3}} + x^{\frac{1}{3}} \cdot 8^{\frac{1}{3}} + 8^{\frac{2}{3}})}{(x-8)}$$

$$\lim_{x \rightarrow 8} x^{\frac{2}{3}} + x^{\frac{1}{3}} \cdot 8^{\frac{1}{3}} + 8^{\frac{2}{3}} = "$$

$$\lim_{x \rightarrow 8} 8^{\frac{2}{3}} + 8^{\frac{1}{3}} \cdot 8^{\frac{1}{3}} + 8^{\frac{2}{3}} =$$

$$3 \cdot 8^{\frac{2}{3}} = 3 \cdot (2^3)^{\frac{2}{3}} =$$

$$3 \cdot 2^{3 \cdot \frac{2}{3}} = 3 \cdot 4 = 12$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt[3]{1+x} - \sqrt[3]{1-x}}$$

$$\begin{aligned} & \left((1+x)^{1/3} - (1-x)^{1/3} \right) \left((1+x)^{2/3} + (1+x)^{1/3}(1-x)^{1/3} + (1-x)^{2/3} \right) \\ & (a^{1/3} - b^{1/3}) (a^{2/3} + a^{1/3}b^{1/3} + b^{2/3}) = \\ & (a - b) = 1+x - (1-x) = \\ & 1+x-1+x = 2x \end{aligned}$$

$$\pi(P) = QP - TC =$$

$$(500 - 5P)P - (200 + 10Q)$$

$$= (500 - 5P)P - (200 + 10(500 - 5P))$$

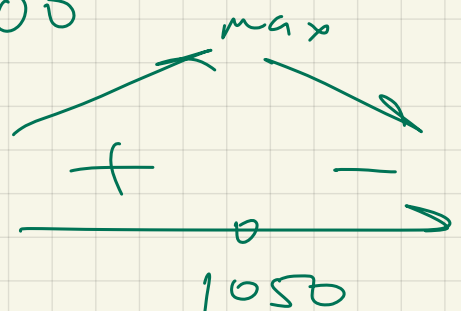
$$\pi(P) = \underline{-5P^2} + 500P - 200 + 5000 - 10000$$

$$\pi'(P) = \underline{500 - 10P} + 5 \cdot 2000 \cdot 10 = 0$$

$$10P = 500 + 10000$$

$$10P = 10500$$

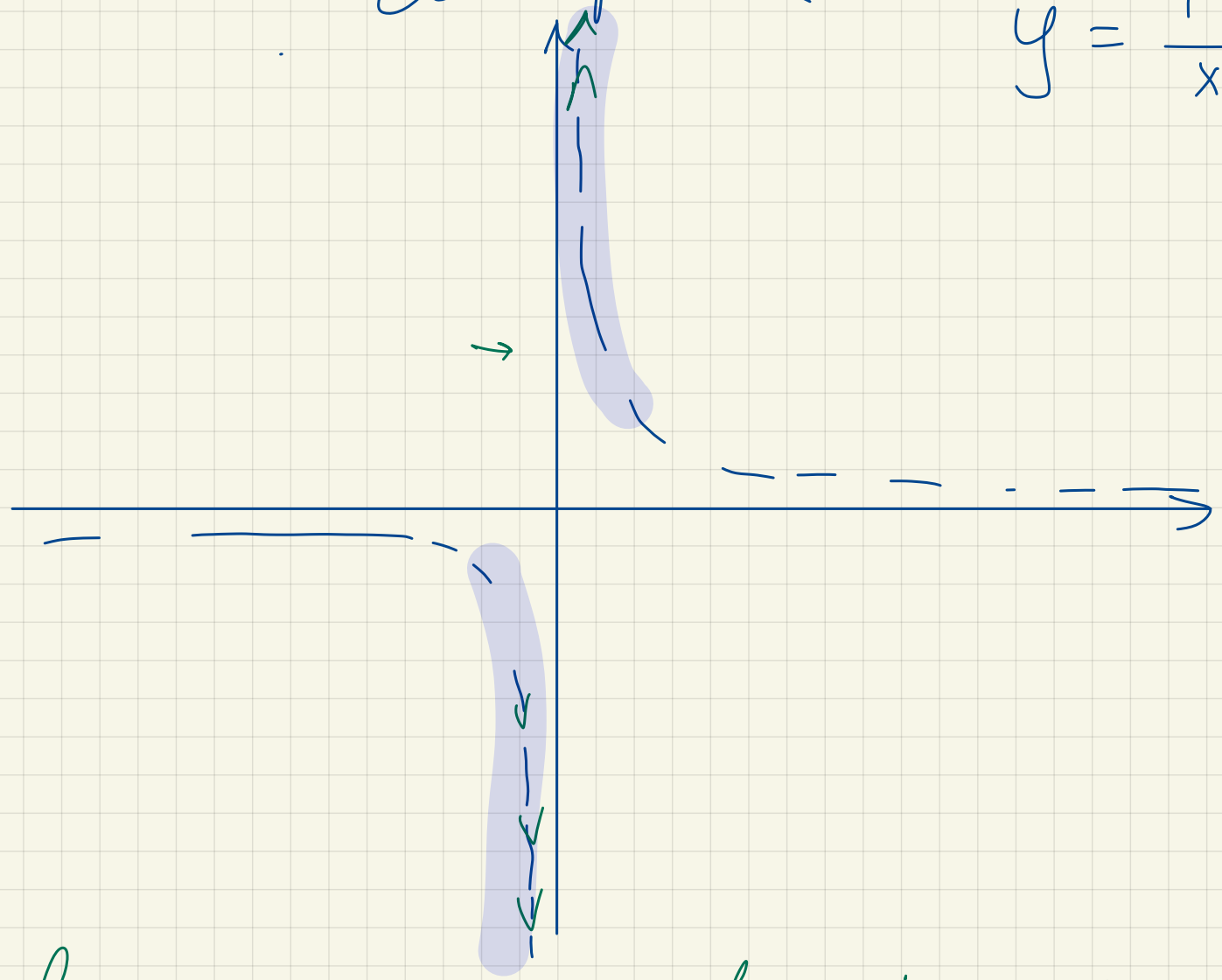
$$\underline{P = 1050}$$



$$\star(1050) =$$

Гипербола

$$y = \frac{1}{x}$$



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = +0$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

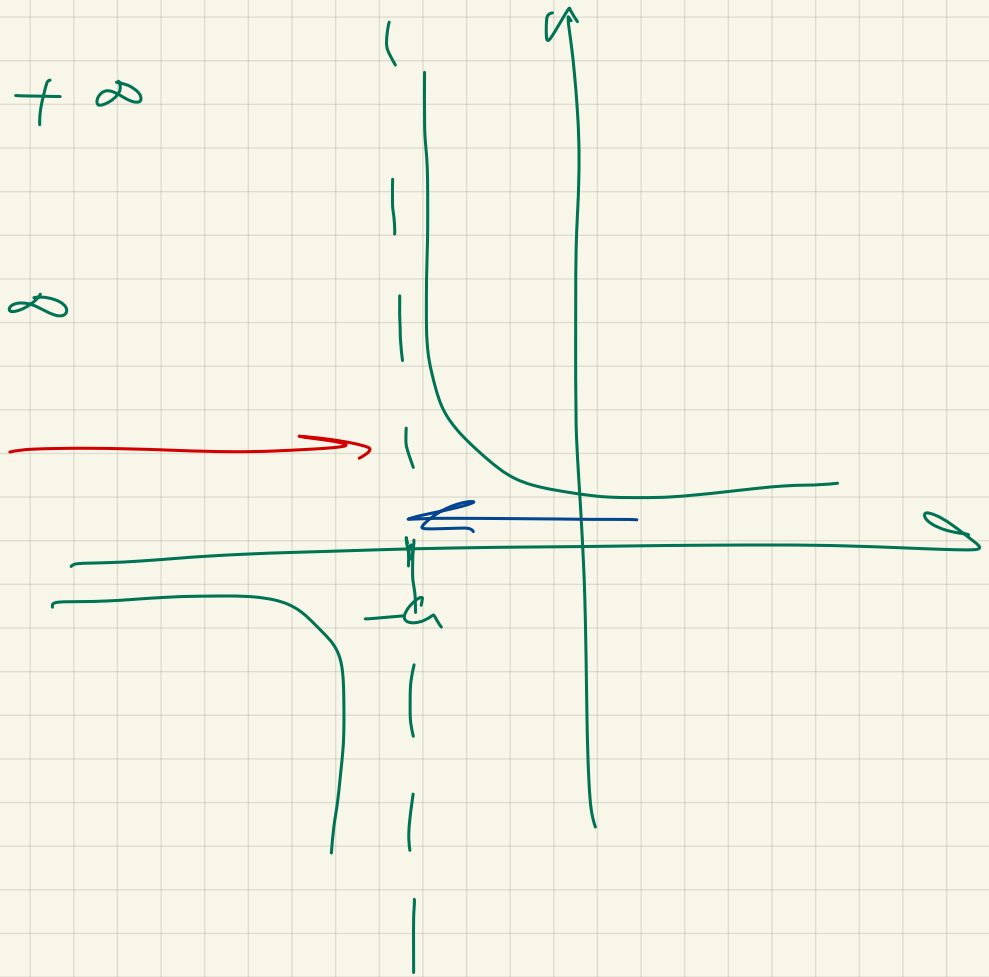
$$\lim_{x \rightarrow -\infty} \frac{1}{x} = -0$$

$$y = \frac{1}{x+a}$$

$$x \neq -a$$

$$\lim_{x \rightarrow -a^+} \frac{1}{x+a} = +\infty$$

$$\lim_{x \rightarrow -a^-} \frac{1}{x+a} = -\infty$$



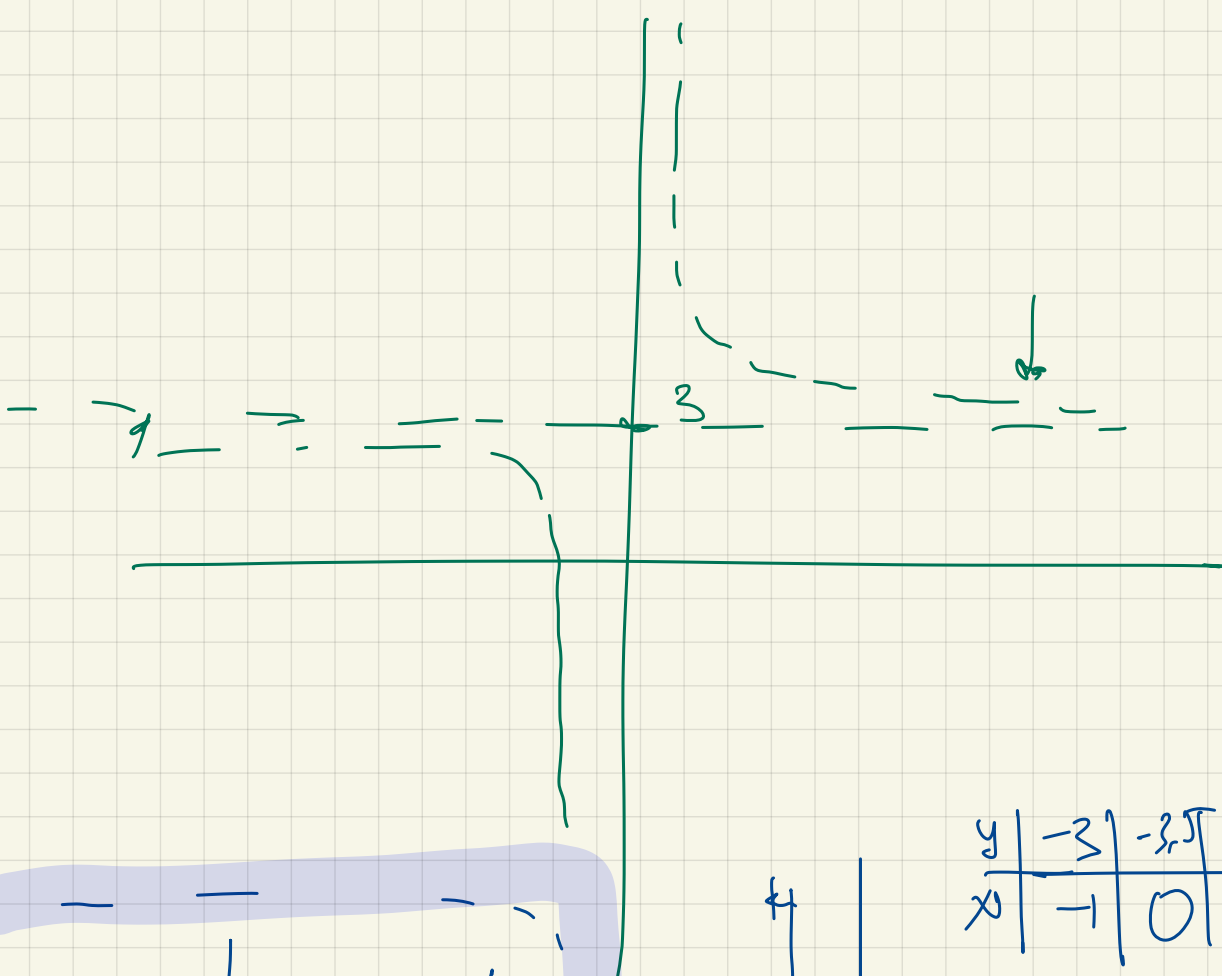
$$y = \frac{1}{x} + 3$$

$$\lim_{x \rightarrow +0} \frac{1}{x} + 3 = +\infty$$

$$\lim_{x \rightarrow -0} \frac{1}{x} + 3 = -\infty$$

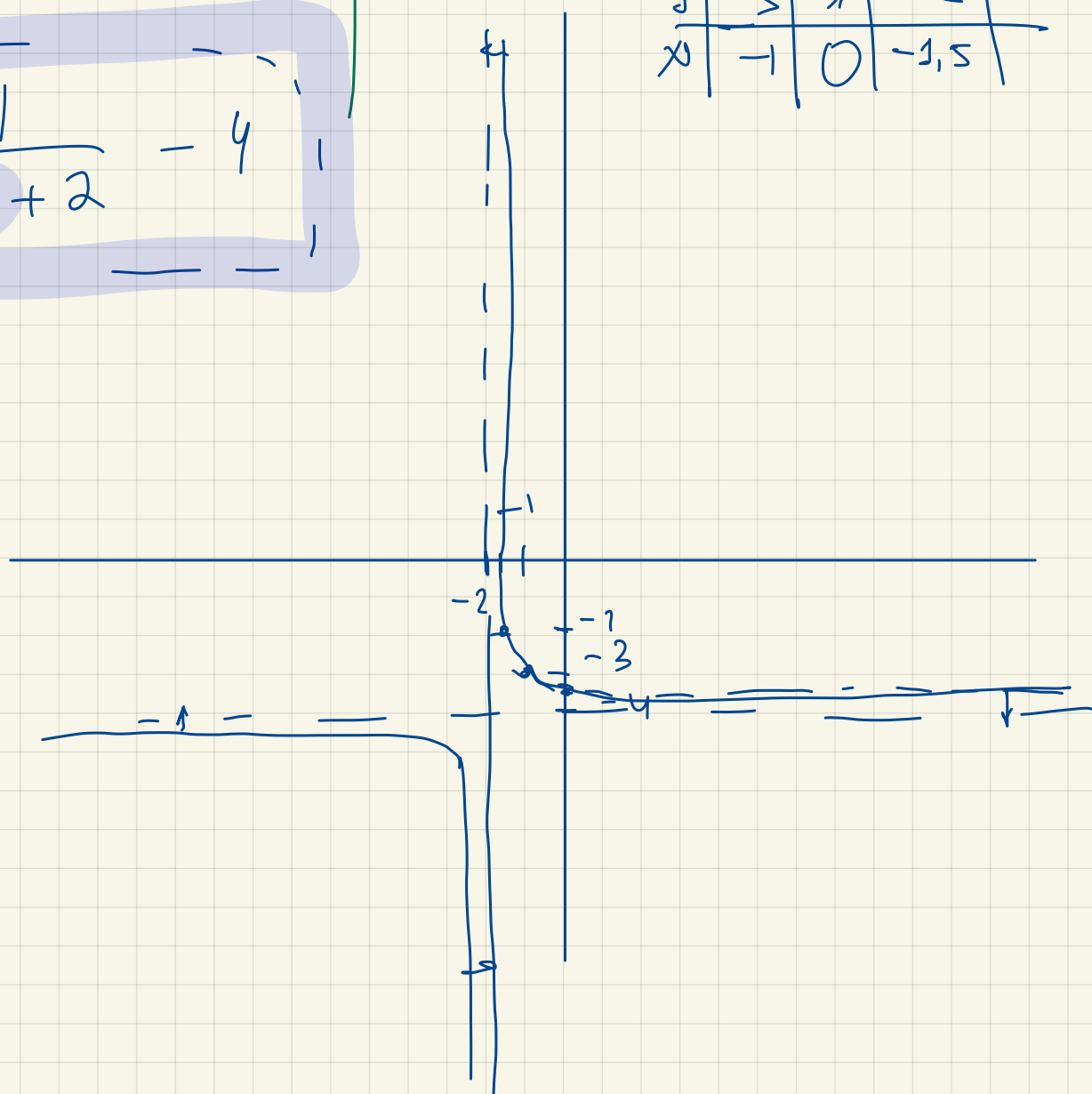
$$\lim_{x \rightarrow +\infty} \frac{1}{x} + 3 = +3^+$$

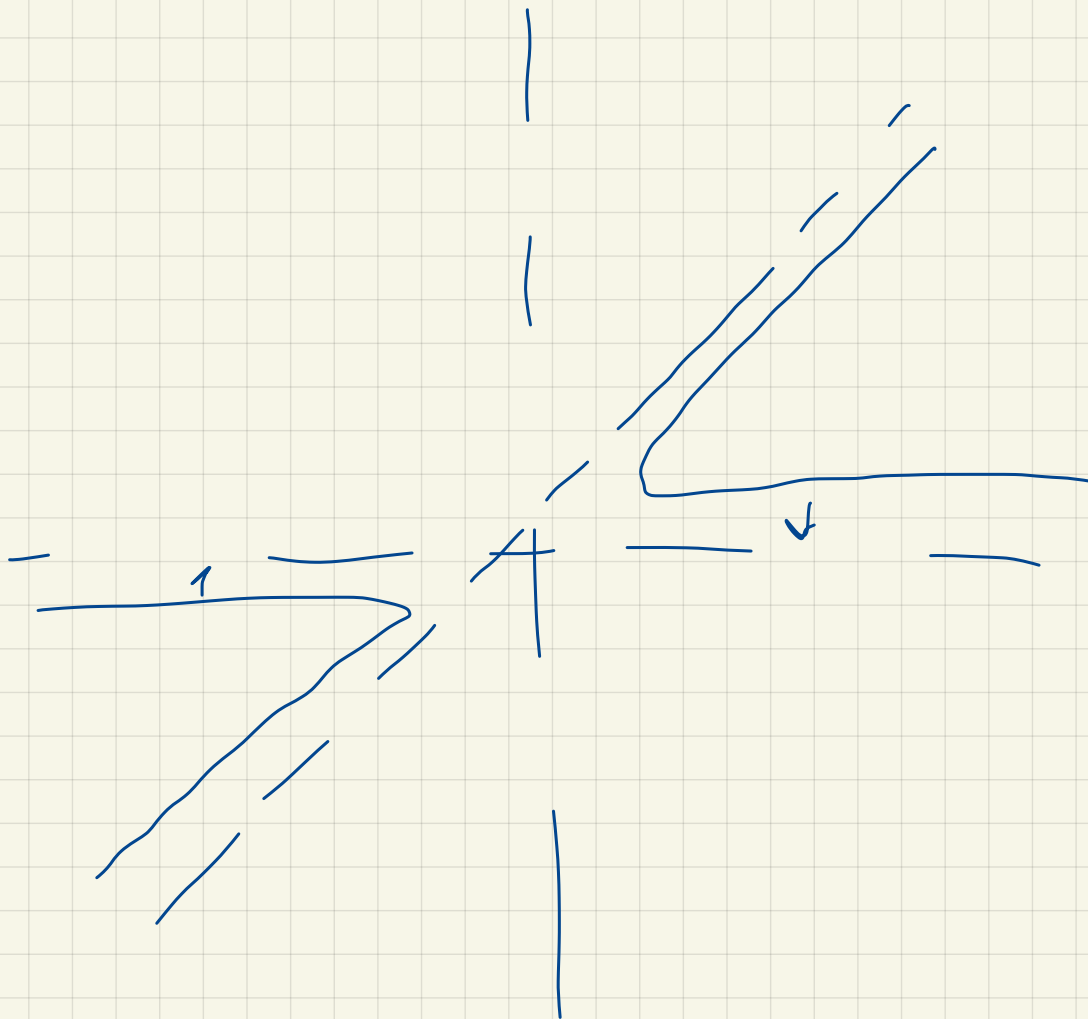
$$\lim_{x \rightarrow -\infty} \frac{1}{x} + 3 = +3^-$$



$$y = \frac{1}{x+2} - 4$$

y	-3	-3,5	-2
x	-1	0	-1,5





$$y = \textcircled{k}x + \textcircled{b}$$

$$k = \lim_{x \rightarrow \infty} \frac{f(x)}{x}$$

$$b = \lim_{x \rightarrow \infty} (f(x) - kx)$$

$$y = \frac{2x^2 + 3x}{x - 1}$$

$$x = 1$$

$$\textcircled{K} = \lim_{x \rightarrow \infty} \left(\frac{2x^2 + 3x}{x-1} \right) \frac{1}{x} =$$

$$= \lim_{x \rightarrow \infty} \frac{2x+3}{x-1} =$$

$$\frac{2(x-1) + 2 + 3}{x-1} = 2 + \frac{5}{x-1}$$

$$2x+3 = 2x - \underline{2} + \underline{2} + 3$$

$$2x-2 = 2(x-1)$$

$$\frac{2(x-1) + 5}{x-1} = 2 + \frac{5}{x-1}$$

$$\lim_{x \rightarrow \infty} 2 + \lim_{x \rightarrow \infty} \frac{5}{x-1} = \textcircled{2}$$

$\downarrow$ $\downarrow$
 2 0

$$b = \lim_{x \rightarrow \infty} \left(\frac{2x^2 + 3x}{x-1} - 2x \right)$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{2x^2} + 3x - \cancel{2x^2} + 2x}{x-1} =$$

$$\lim_{x \rightarrow \infty} \frac{5x}{x-1} = \lim_{x \rightarrow \infty} \frac{5x - 5 + 5}{x-1}$$

$$\lim_{x \rightarrow \infty} \frac{5(x-1) + 5}{x-1} = \lim_{x \rightarrow \infty} \frac{5}{1} + \lim_{x \rightarrow \infty} \frac{5}{x-1}$$

$$k=2$$

$$b=5$$

